



Rossmoyne Senior High School

Semester One Examination, 2020

Question/Answer booklet

**MATHEMATICS
METHODS
UNIT 1**

**Section Two:
Calculator-assumed**

SOLUTIONS

WA student number: In figures

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In words

Circle your Teacher's Name: **ALVARO** **BESTALL** **FRASER-JONES** **GOH**
 KOULIANOS **LUZUK** **RUDLAND** **TANDAY**

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up
to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

See next page

Structure of this paper

	Number of questions available	Number of questions to be attempted	Working time (minutes)	Marks available	Percentage of exam
Section One Calculator—free	9	9	50	54	36
Section Two Calculator—assumed	15	15	100	96	64
				150	100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2020*. Sitting this examination implies that you agree to abide by these rules.
2. Answer the questions according to the following instructions.

Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

It is recommended that you **do not use pencil**, except in diagrams.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.
5. The Formula Sheet is **not** handed in with your Question/Answer Booklet.

Section Two: Calculator–assumed

64% (96 marks)

This section has **fifteen (15)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Working time: 100 minutes

Question 10 (6 marks)

- (a) A diameter of a circle has endpoints $P(-4,0)$ and $Q(16,0)$. Determine the y – intercepts of the circle. (3 marks)

Solution	
Midpoint of PQ (6, 0)	$(x - 6)^2 + y^2 = 100$
y – intercepts	(0, 8) or (0, -8)
Specific behaviours	
• ✓ midpoint • ✓ equation of circle • ✓ correct y-intercepts	

- (b) The circumference of a circle (C) is directly proportional to the radius (r).

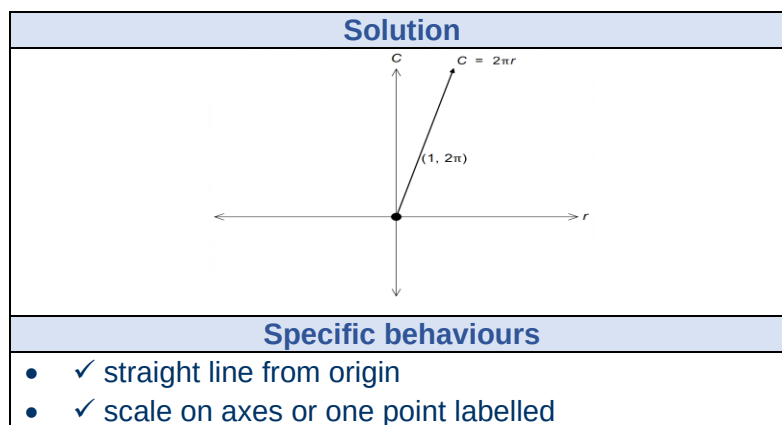
- (i) State the exact constant of proportionality.

(1 mark)

Solution
$k = 2\pi$
Specific behaviours
• ✓ correct answer

- (ii) Draw a graph on the axes below to show the relationship between C and r . Label the axes and all critical features.

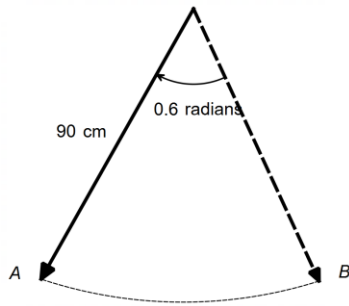
(2 marks)



See next page

Question 11 (6 marks)

A pendulum is 90 cm long and swings through an angle of 0.6 radians. The extreme positions of the pendulum are indicated by the points A and B in the diagram.



- (a) Find the length of the arc AB.

(2 marks)

Solution
$l = r\theta = (90)(0.6)$ $l = 54 \text{ cm}$
Specific behaviours
✓ correct substitution ✓ correct value

- (b) Find the straight line distance between the extreme positions of the pendulum (i.e. the distance from A to B.). Round to 3 decimal places.

(2 marks)

Solution
$AB^2 = 90^2 + 90^2 - 2(90)(90)\cos 0.6$ $AB = 53.194 \text{ cm}$
Specific behaviours
✓ correct substitution ✓ correct value

- (c) Find the area of the sector swept out by the pendulum.

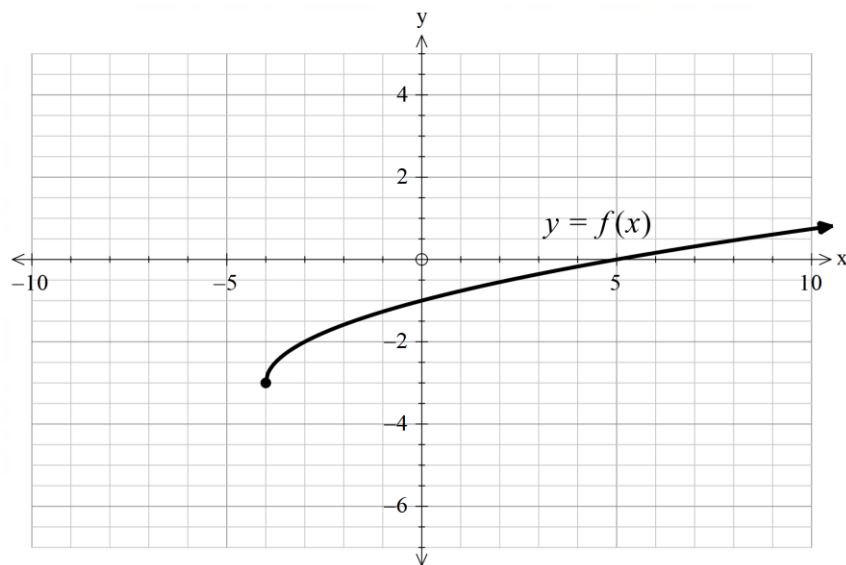
(2 marks)

Solution
$A = \frac{1}{2} r^2 \theta = \frac{1}{2} (90)^2 (0.6)$ $A = 2430 \text{ cm}^2$
Specific behaviours
✓ correct substitution ✓ correct value

Question 12 (4 marks)

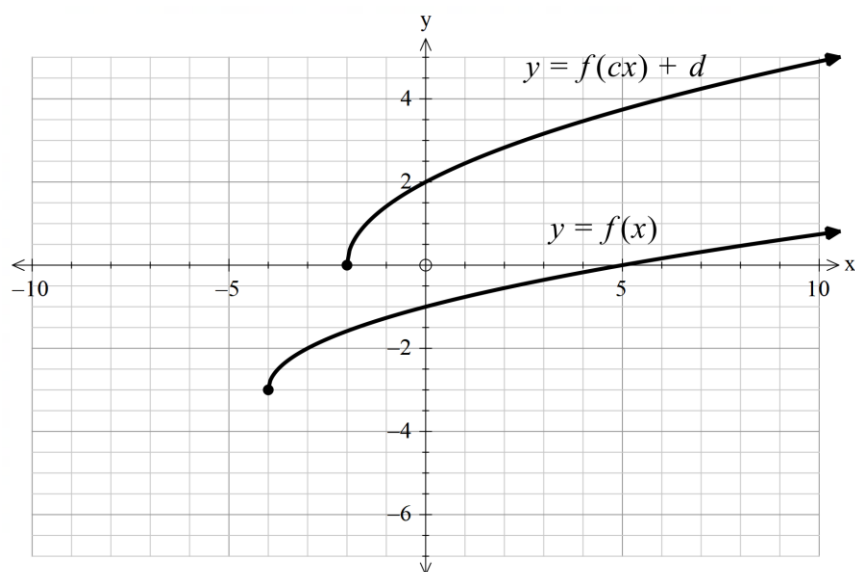
The graph of $y = f(x)$ is drawn below, where $f(x) = \sqrt{x - a} + b$.

- (a) Determine the value of the constant a and the value of the constant b . (2 marks)



Solution
$a = -4$ $b = -3$
Specific behaviours
✓ correct a value ✓ correct b value

- (b) The graph of $y = f(cx) + d$ is now graphed on the same axes

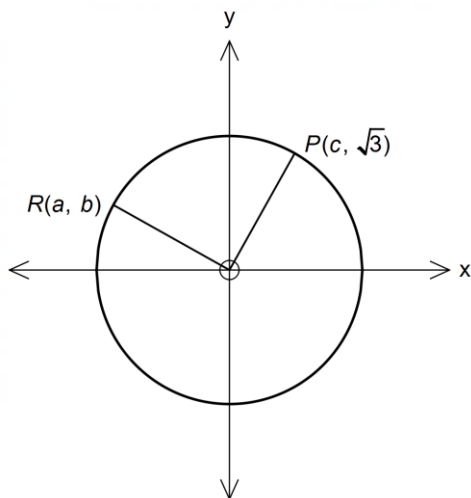


Solution
$c = 2$ $d = 3$
Specific behaviours
✓ correct c value ✓ correct d value

- Determine the value of the constant c and the value of the constant d . (2 marks)

Question 13 (7 marks)

The radius of a circle, with centre O , is 2 units. $R(a, b)$ and $P(c, \sqrt{3})$ are points on the circle. Let the angle between the positive x – axis and the line OP be θ and the angle ROP is 90° .



- (a) Show that $c = 1$ (2 marks)

Solution
$2^2 - (\sqrt{3})^2 = c^2 \therefore c = 1$

behaviours
✓ Appropriate method (trig or pythag) ✓ Subs to find c

- (b) Determine the value(s) of:

- (i) $\sin \theta$ and hence θ . (2 marks)

Solution
$\sin \theta = \frac{\sqrt{3}}{2} \therefore \theta = 60^\circ \text{ (or } \frac{\pi}{3} \text{ radians)}$
Specific behaviours
✓ correct value of $\sin \theta$ ✓ correct value of θ

- (ii) $\tan(360^\circ - \theta)$. (1 mark)

Solution
$-\tan \theta = -\sqrt{3} \text{ or } -1.732$

- (iii) a and b . (2 marks)

Solution
$a = -\sqrt{3} \quad b = 1$
Specific behaviours
✓ correct value of a ✓ correct value of b

Question 14 (6 marks)

If it rains on a given day, the probability that it rains the next day is $\frac{1}{3}$. If it does not rain on a given day, the probability that it rains the next day is $\frac{1}{6}$. The probability that it will rain tomorrow is $\frac{1}{5}$.

- (a) Draw a tree diagram to show all the probabilities and outcomes. (3 marks)

Solution	
Outcomes	
Rain	RR
No rain	$R\bar{R}$
Rain	$\bar{R}R$
No rain	$\bar{R}\bar{R}$
Specific behaviours	
✓ correct label and probabilities day 1 ✓ correct label and probabilities day 2 ✓ correct outcomes listed	

- (b) Determine the probability that:

- (i) it will rain tomorrow and the next day. (1 mark)

Solution
$\frac{1}{15}$

- (ii) it will rain the day after tomorrow. (1 mark)

Solution
$\left(\frac{1}{5} \times \frac{1}{3}\right) + \left(\frac{4}{5} \times \frac{1}{6}\right) = \frac{3}{15} = \frac{1}{5}$

accept $\frac{3}{15}$

- (iii) there will be no rain the next day, given that it rains tomorrow. (1 mark)

Solution
$\frac{2}{3}$

Question 15 (9 marks)

- (a) Line q is perpendicular to the line with the equation $y = 2x - 3$. Line q has the same x -intercept as the line with the equation $x + 2y = 6$. Determine the equation of line q .

(3 marks)

Solution	
$m = -\frac{1}{2} \therefore y = -\frac{1}{2}x + c$	
x-intercept (6, 0)	$0 = -\frac{1}{2}(6) + c$
$c = 3$	$y = -\frac{1}{2}x + 3$
Specific behaviours	
✓ correct gradient ✓ substitute (6,0) ✓ correct value of c and equation	

- (b) For what value(s) of k is the line through the points $(3, 2k + 1)$ and $(8, 4k - 5)$ parallel to the x -axis. (2 marks)

Solution
$\frac{(2k + 1) - (4k - 5)}{3 - 8} = 0$ $-2k + 6 = 0 \therefore k = 3$
Specific behaviours
✓ equate gradient expression to 0 ✓ solve and state value of k

Alt Solution
$2k + 1 = 4k - 5$ $k = 3$
Specific behaviours
✓ equate two Y-coordinates ✓ solves and state value of k

- (c) Consider the quadratic equation $x^2 - (p + 7)x + p + 87 = 0$ where p is a negative real number. Solve for x given that the equation has one distinct real solution. (4 marks)

Solution
$b^2 - 4ac = 0 \quad (-p - 7)^2 - 4(1)(p + 87) = 0$ $p = -23 \text{ or } 13$ $x^2 - (p + 7)x + p + 87 = 0 \text{ where } p = -23$ $\therefore x^2 + 16x + 64 = 0 \quad x = -8$
Specific behaviours
✓ correct substitution into discriminant ✓ correct value of p ✓ correct quadratic equation in x ✓ solve and state correct x value No penalty for substituting $(p + 7)$ instead of $(-p - 7)$

See next page

Question 16 (4 marks)

It is known that for events A and B, $P(\bar{A}) = 0.6$, $P(B) = 0.7$ and $P(A|B) = 0.5$

- (a) Use this information to complete the two-way table.

(3 marks)

	$P(A)$	$P(\bar{A})$	Total
$P(B)$	0.35	0.35	0.7
$P(\bar{B})$	0.05	0.25	0.3
Total	0.4	0.6	1

Solution
See table
Specific behaviours
✓ correct $A \cap B$ ✓ correct 0.6 and 0.7 ✓ balance correct

- (b) Hence or otherwise, determine $P(\bar{A} \cap \bar{B})$

(1 mark)

Solution
$P(\bar{A} \cap \bar{B}) = 0.25$
Specific behaviours
✓ correct number

Question 17 (4 marks)

The probabilities that Alf and Bess will attend the next work social are $P(A) = 0.7$, $P(B) = 0.55$ respectively. It is also known that $P(A \cap B) = 0.45$.

- (a) Determine $P(A \cup B)$.

(1 marks)

Solution
$P(A \cup B) = 0.7 + 0.55 - 0.45$ $= 0.8$
Specific behaviours
✓ correct number

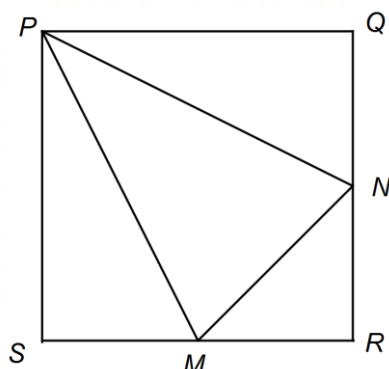
- (b) Describe, in the context of this question, the event $(A \cap \bar{B}) \cup (\bar{A} \cap B)$ and calculate the probability that it happens.

(3 marks)

Solution
Only one of Alf or Bess would attend the social $P = 0.25 + 0.1 = 0.35$
Specific behaviours
✓ correct explanation ✓ correct substitution ✓ correct probability

Question 18 (5 marks)

In the diagram, square PQRS has side length 2 cm. Points M and N are midpoints of SR and RQ respectively.



(a) Show that the angle MPN = 0.6435 radians.

(3 marks)

Solution
Pythagoras $PM = \sqrt{5}$ $PN = \sqrt{5}$ $MN = \sqrt{2}$ $2 = 5 + 5 - 2(5)\cos\theta$ $\theta = 0.6435$
Specific behaviours
✓ pythagoras ✓ Cos rule ✓ correct answer

Alt Solution
trig $\angle SPM = \tan^{-1}(1/2) = 0.4636$ $\angle QPN = \tan^{-1}(1/2) = 0.4636$ $\angle MPN = \frac{\pi}{2} - 2(0.4636)$ $= 0.6435$
Specific behaviours
✓ $\angle SPM$ and $\angle QPN$ ✓ expression for $\angle MPN$ ✓ correct answer

(b) Hence, or otherwise determine the area of ΔMPN .

(2 marks)

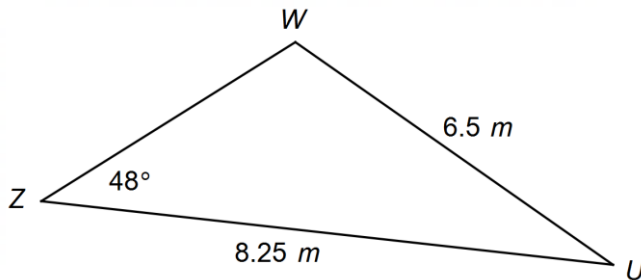
Solution
$\text{Area} = \frac{1}{2}(5)\sin(0.6435) = 1.5 \text{ cm}^2$ Or $A = \frac{1}{2} \cdot 2.1 \times \frac{1}{2} \cdot 1.1$ $= 4 - 2.5$ $= 1.5 \text{ cm}^2$
Specific behaviours
✓ correct substitution ✓ correct answer

Question 19 (7 marks)

- (a) Three aircraft, marked, Z, U and W, are flying in a triangular formation at an air show, at the same speed and altitude. Aircraft Z is 8.25 m from aircraft U which is 6.5 m from aircraft W. The angle WZU formed is 48° .

Determine what distance aircraft W is from aircraft Z?

(3 marks)

Solution

Soln 1

$$\frac{\sin W}{8.25} = \frac{\sin 48}{6.5} \quad \angle W = 70.6^\circ \text{ or } 109.4^\circ$$

$$8.25^2 = x^2 + 6.5^2 - 2(6.5)x\cos(70.6) \quad x = 7.68 \text{ m}$$

$$8.25^2 = x^2 + 6.5^2 - 2(6.5)x\cos(109.4) \quad x = 3.36 \text{ m}$$

Aircraft W could be 7.68 m or 3.36 m from aircraft Z

Or

Soln 2

$$\text{Solve } 6.5^2 = 8.25^2 + x^2 - 2 \cdot x \cdot (8.25) \cos 48^\circ$$

$$x = 3.36 \text{ m} \quad \text{or} \quad 7.68 \text{ m}$$

Specific behaviours

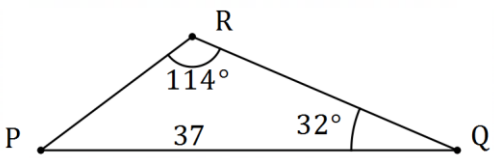
- ✓ sine rule and both solns
- ✓ cos rule and first x value
- ✓ second x value

or second soln

- ✓ cos rule
- ✓ first x value
- ✓ second x value

Only one correct solution 2/3

- (b) Determine the area of triangle PQR when $\angle PQR = 32^\circ$, $\angle PRQ = 114^\circ$ and $PQ = 37$ cm.
(4 marks)

Solution	
	
$\frac{q}{\sin 32^\circ} = \frac{37}{\sin 114^\circ} \Rightarrow q = 21.46$	
$\begin{aligned}\text{Area} &= \frac{1}{2} (37)(21.46) \sin(34^\circ) \\ &= 222 \text{ cm}^2\end{aligned}$	
Specific behaviours	
• ✓ sketch of triangle • ✓ correct use of sine rule • ✓ length of second side q • ✓ correct area This is our unit question – subtract one if units are not provided in the final answer	

Question 20 (8 marks)

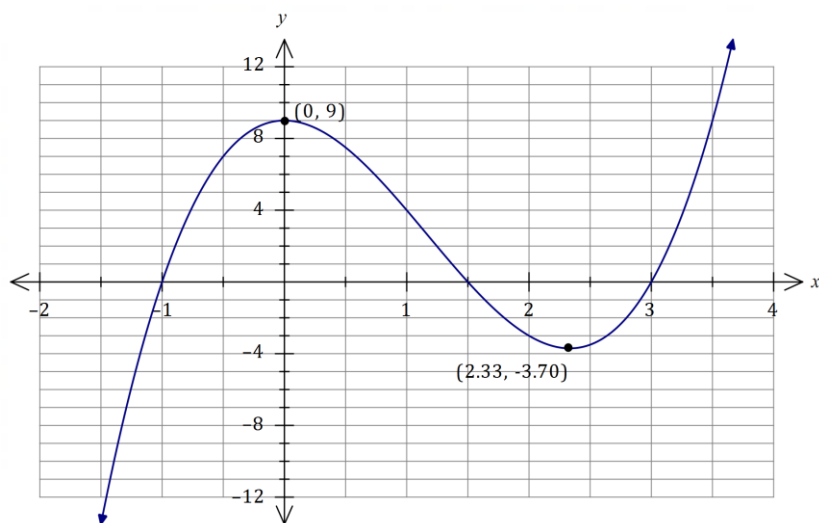
A polynomial of degree 3 passes through the points with coordinates $(0, 9)$, $(-1, 0)$, $(3, 0)$ and $(1.5, 0)$.

- (a) Determine the equation of the polynomial. State in expanded form. (3 marks)

Solution
Using roots: $y = a(x + 1)(x - 3)(x - 1.5)$
Use 4th point: $x = 0 \Rightarrow 9 = a(1)(-3)(-1.5)$ $a = 2$
Expand: $y = 2(x + 1)(x - 3)(x - 1.5)$ $= 2x^3 - 7x^2 + 9$
Specific behaviours
✓ factored form using roots ✓ substitutes fourth point and determines correct value of a ✓ correct expanded form

Alt Solution
Using roots: $y = b(x + 1)(x - 3)(2x - 3)$
Use 4th point: $x = 0 \Rightarrow 9 = b(1)(-3)(-3)$ $b = 1$
Expand: $y = (x + 1)(x - 3)(2x - 3)$ $= 2x^3 - 7x^2 + 9$
Specific behaviours
✓ factored form using roots ✓ substitutes fourth point and determines correct value of b ✓ correct expanded form

- (b) Draw the graph of the polynomial on the axes below, indicating and labelling key features (4 marks)



Solution
See graph
Specific behaviours
✓ all roots ✓ labelled maximum ✓ labelled minimum ✓ smooth curve

- (c) Determine the coordinates of the point where concavity changes. (1 mark)
 Give your answer to an accuracy of 4 decimal places.

Solution
$(1.6667, 2.6481)$
Specific behaviours
✓ correct answer

Question 21 (6 marks)

Two events are such that $P(X) = 0.2$, $P(Y) = 0.5$ and $P(Y|X) = 0.1$.

Determine the probability that

- (a) both events occur.

(2 marks)

Solution
$P(Y X) = \frac{P(X \cap Y)}{P(X)} \Rightarrow P(X \cap Y) = 0.2 \times 0.1 = 0.02$
Specific behaviours
✓ indicates use of conditional formula ✓ correct probability

- (b) at least one event occurs.

(2 marks)

Solution
$\begin{aligned} P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\ &= 0.2 + 0.5 - 0.02 \\ &= 0.68 \end{aligned}$
Specific behaviours
✓ indicates use of rule ✓ correct probability

- (c) neither event occurs.

(1 mark)

Solution
$\begin{aligned} P(X \cap Y) &= 1 - P(X \cup Y) \\ &= 1 - 0.68 \\ &= 0.32 \end{aligned}$
Specific behaviours
✓ correct probability

- (d) X occurs given that Y has occurred.

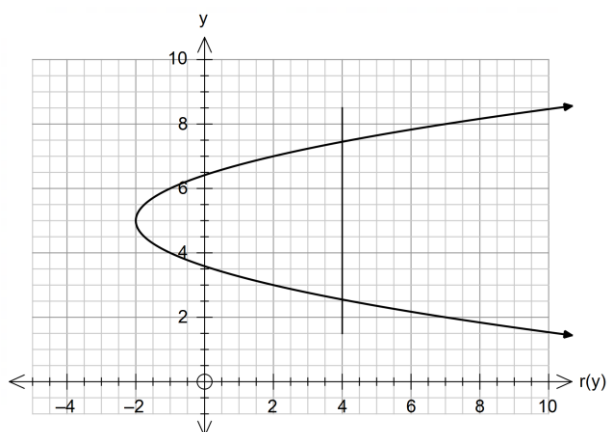
(1 mark)

Solution
$\begin{aligned} P(X Y) &= \frac{P(X \cap Y)}{P(Y)} \\ &= \frac{0.02}{0.5} \\ &= 0.04 \end{aligned}$
Specific behaviours
✓ correct probability

Question 22 (6 marks)

(a) (i) State the equation of the graph below:

(2 marks)



Solution	
$x + 2 = (y - 5)^2$ or $f(y) = x = (y - 5)^2 - 2$ or $y = \pm\sqrt{x + 2} + 5$	
Specific behaviours	
• ✓✓ correct equation	
Part marks (only one of ..)	
• ✓ correct x component	$x + 2$
• ✓ correct y component	$y - 5$

(ii) Apply the vertical line test to the graph above and explain its significance. (2 marks)

Solution
The vertical line intersects the graph at two points Therefore this relation is not a function
Specific behaviours
• ✓ Show a vertical line on the graph • ✓ explanation

(b) A straight line makes an angle of 60° with the positive x -axis and passes through the point with coordinates $(0, -6)$. Determine the exact equation of the line. (2 marks)

Solution
$m = \tan 60^\circ = \sqrt{3}$ $y = \sqrt{3}x - 6$
Specific behaviours
• ✓ gradient • ✓ equation in exact form

Question 23 (9 marks)

The manager of a private bus company has worked out a model for the relationship between the number of passengers carried each week and the profit, (in tens of \$), the company makes.

If n is the number of passengers carried, the profit is given by $P(n) = 2n - 2000 - \left(\frac{n}{100}\right)^2$.

- (a) Can the relation $P(n)$ be described as a function. Justify your answer. (1 mark)

Solution
Yes, every value of n has only one value of P . (one-to-one mapping)
Or as a quadratic – it is a function
Specific behaviours
• ✓ correct answer with justification

- (b) Determine the minimum number of passengers needed per week for the company to make a profit. (1 mark)

Solution
Solve $P(n)=0$ CAS soln = 1055.7 $N=1056$ passengers to make a profit
Specific behaviours
• ✓ correct answer

- (c) Determine the maximum profit made according to this model and how many passengers are carried during the week to achieve this profit. (2 marks)

Solution
Coordinates of Max TP (10000, 8000) CAS 10000 passengers and \$80 000 profit for that week
Specific behaviours
• ✓ correct number of passengers • ✓ correct profit in dollars

- (d) (i) Determine the simplified expression for $P(n + 1) - P(n)$, and explain what this expression represents. (4 marks)

Solution	
Define $P(n)$	CAS
$P(n + 1) - P(n) = 2(n + 1) - 2000 - \left(\frac{n + 1}{100} \right)^2 - \left(2n - 2000 - \left(\frac{n}{100} \right)^2 \right)$	
Simplify $[P(n+1)-P(n)] = -\frac{n}{5000} + 1.9999$ CAS	
Represents the increase in profit if one extra passenger is carried.	
Specific behaviours	
✓ ✓ correct expression ✓ simplified expression ✓ correct explanation	

- (ii) How many passengers would the bus company carry in a week if the profit made by one extra passenger was 50 cents? (2 marks)

Solution		
Solve $\left[-\frac{n}{5000} + 1.9999 = 0.50 \right]$	CAS	(soln= 7499.5)
$n = 7500$ passengers to make a profit		
Specific behaviours		
✓ equates expression to 0.5 , and solves ✓ correct answer		

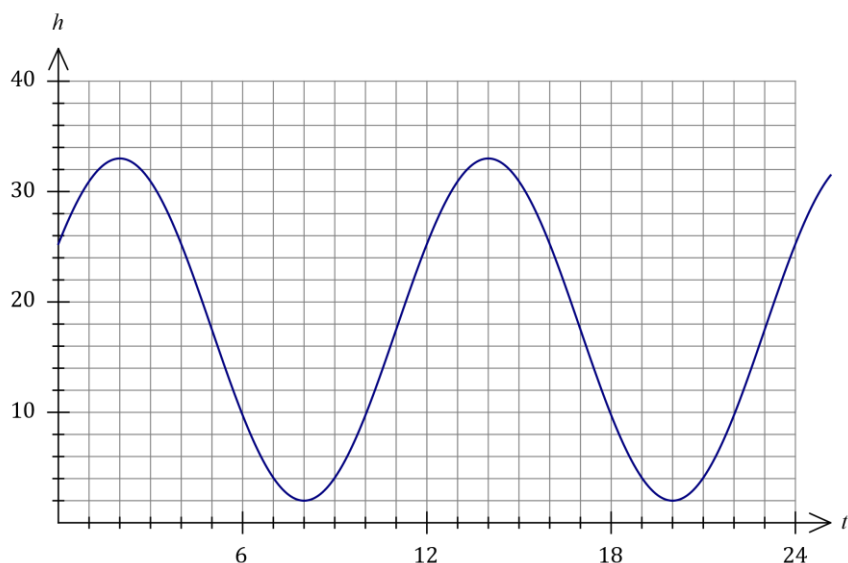
Question 24 (8 marks)

The height above ground level, h m, of a seat on a steadily rotating Ferris wheel t minutes after the wheel begins to move is given by $h = 17.5 + 15.5 \cos\left(\frac{\pi t}{6} - \frac{\pi}{3}\right)$.

- (a) Determine the initial height of the seat. (1 marks)

Solution
$h(0) = 25.25$ m
Specific behaviours
• ✓ correct height

- (b) Graph the height of the seat against time on the axes below. (4 marks)



Solution
• See graph
Specific behaviours
• ✓ (0, 25) and (24, 25)
• ✓ correct maximums
• ✓ correct minimums
• ✓ smooth curve

- (c) Determine

- (i) the maximum height above ground reached by the seat. (1 mark)

Solution
• $h_{MAX} = 33$ m
Specific behaviours
• ✓ correct height

- (ii) the time taken, to the nearest second, for the seat to first reach a height of 7 m above ground level. (2 marks)

Solution
• $h = 7 \Rightarrow t = 6.42$
• $0.42 \times 60 = 25$
• $t = 6 \text{ m } 25 \text{ s (385 s)}$
Specific behaviours
• ✓ time as decimal
• ✓ time to nearest second

Additional working space

Question number(s):

Additional working space

Question number(s):